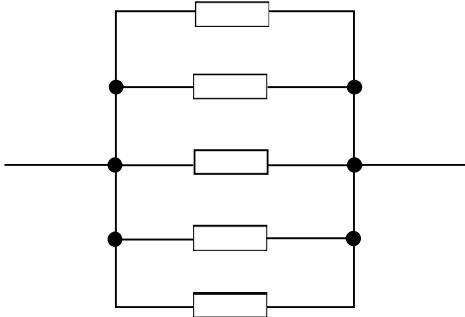


	$\eta = 3.35 \times 10^{-11} \text{ m}$
4(a)	Ω $= \text{V A}^{-1} = \text{J C}^{-1} \text{ A}^{-1} = \text{N m C}^{-1} \text{ A}^{-1}$ $= (\text{kg m s}^{-2}) \text{ m (A}^{-1} \text{ s}^{-1}) \text{ A}^{-1}$ $= \text{kg m}^2 \text{ s}^{-3} \text{ A}^{-2}$
4(b)(i)	$P = I^2 R = I^2 \frac{\rho L}{A}$ $P_A = (25)^2 \frac{(1.68 \times 10^{-8})(8.0)}{4.00 \times 10^{-6}} = 21 \text{ W}$ $P_B = (25)^2 \frac{(2.65 \times 10^{-8})(12.0)}{6.00 \times 10^{-6}} = 33.1 \text{ W}$
4(b)(ii)	$\frac{V}{L} = \frac{IR}{L} = \frac{I}{L} \frac{\rho L}{A} = \frac{I\rho}{A}$ Wire A: $\frac{V}{L} = \frac{25(1.68 \times 10^{-8})}{4.00 \times 10^{-6}} = 0.105 \text{ V m}^{-1}$ Wire B: $\frac{V}{L} = \frac{25(2.65 \times 10^{-8})}{6.00 \times 10^{-6}} = 0.110 \text{ V m}^{-1}$
4(b)(iii)	Wire A is the most suitable for the connection to the cooker. Wire A has a lower rate at which electrical energy is converted into thermal energy. Therefore there is less danger of fire or burns when using wire A.

No.	Solution
4(c)(i)	
4(c)(ii)	$\frac{1}{0.0458} = \frac{5}{R_{\text{wire}}}$ $R_{\text{wire}} = 0.229 \, \Omega$
4(c)(iii)	If a single thick wire of resistance R is used, the power dissipated in the thick wire is $P = I^2 R$ where I is the current flowing through the thick wire. If a cable made of five thin wires (each wire of resistance $5R$) is used, the effective resistance of the cable is R. The power dissipated in each thin wire is $P_{\text{thin wire}} = \left(\frac{I}{5}\right)^2 (5R) = \frac{1}{5} (I^2 R) = \frac{1}{5} P.$ The power dissipated in each thin wire is less than that in the single thick wire. Therefore it will be safer to use a cable of several thin wires.
5(a)	1